

TASK 6 SUBMISSION

Competitor Demand Prediction Using Job Postings

Competitor Comparison

NVIDIA against Google, Microsoft and Meta

Prepared by	Nayab Khalid
Company specialism	NVIDIA
Task	Task 6 of 10 - Competitor Comparison
Companies compared	NVIDIA 1,745 postings · Google 1,227 · Meta 209 · Microsoft 182
Date	20 September 2026
Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

The result that governs everything else in this document

Run against the shared taxonomy as it stands, NVIDIA appears far more skill-rich than the other three: 72% of its postings match a skill, against 29%, 16% and 5%. That result is an artefact. Run against the taxonomy as it was before I extended it in Task 4, NVIDIA scores 21.2% and Google 22.3% - NVIDIA is behind. The entire advantage comes from 34 skills that I added, chosen to describe NVIDIA's vocabulary. Until the other three extend the taxonomy for their own companies, a four-way skill comparison measures the taxonomy, not the companies.

1 What the four datasets actually are

The four members' datasets are not alike, and no amount of analysis makes them alike. This is the starting position:

Company	Postings	Job titles	Description text	Posting dates
NVIDIA	1,745	yes	none - ToS blocked collection	yes, 12 weekly observations
Google	1,227	yes	responsibilities + qualifications	none in the dataset
Meta	209	yes	yes, median 3,239 characters	yes, 2024
Microsoft	182	yes	yes, median 3,715 characters	empty column

Two consequences follow immediately. **No cross-company time series is possible**: Google has no dates, Microsoft's date column is empty, and Meta's data is from 2024 while NVIDIA's is from 2026. Task 5's hiring-velocity work therefore has no counterpart here. And **skill counts are not comparable**, because extracting from a 3,700-character description finds many more skills than extracting from a six-word title.

1.1 How the comparison is made fair

Everything is computed on two bases and reported separately.

Basis	What it uses	Comparable?
A - titles	Job titles for all four companies, which every dataset has	Yes. The basis for every comparison claim here
B - full text	Descriptions for Google, Meta and Microsoft	No. NVIDIA has no text. Included only to quantify what NVIDIA's data cannot see

All four are matched with the same shared taxonomy through the same matcher, which was factored into scripts/skill_matcher.py for this task so that Tasks 4, 6 and 8 cannot drift apart. Google's 23 YouTube rows are excluded; this is a company-level comparison.

2 The validity control

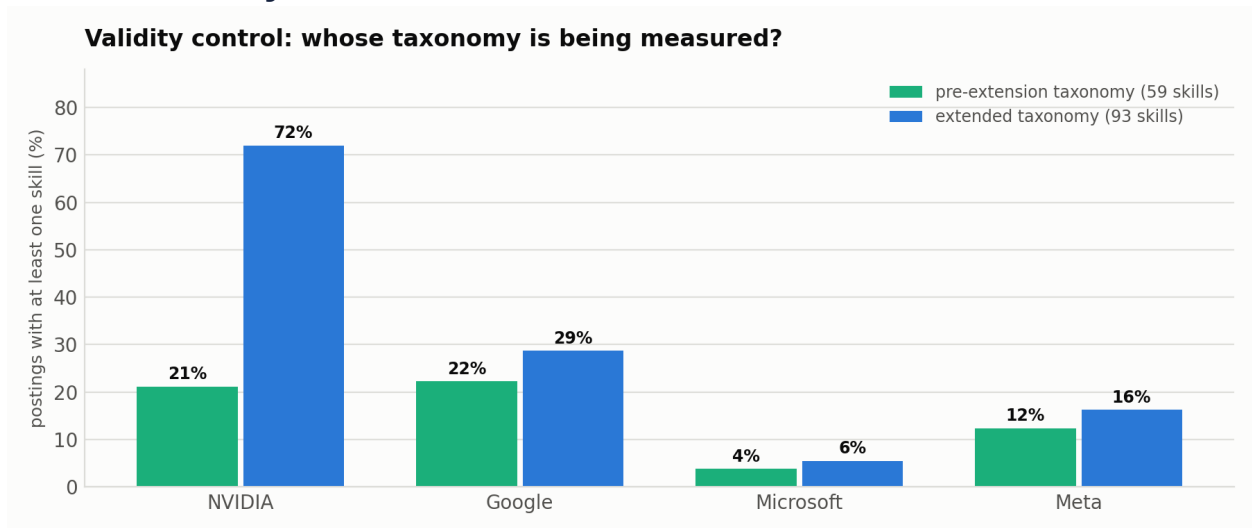


Figure 1. The same job titles, the same matcher, two versions of the taxonomy.

Company	Pre-extension taxonomy (59 skills)	Extended taxonomy (93 skills)	Change
NVIDIA	21.2%	72.0%	+50.8 pts
Google	22.3%	28.7%	+6.4 pts
Meta	12.4%	16.3%	+3.9 pts
Microsoft	3.8%	5.5%	+1.7 pts

Under the taxonomy the team actually agreed, NVIDIA and Google are level and NVIDIA is marginally behind. Under my extension NVIDIA gains 50.8 points and the others gain between 1.7 and 6.4. The extension was honest work - it raised NVIDIA's coverage from 21% to 72% and the reasoning is documented in Task 4 - but it was built by reading NVIDIA titles, so it finds NVIDIA vocabulary.

What this means for the team, concretely

The fix is not to discard the extension. It is for each member to extend the taxonomy for their own company, from their own candidate terms, and then re-run. Until that happens the only unbiased four-way comparison is the 59-skill one, and that one is too sparse to say much: Microsoft matches 3.8% of postings, which is 7 postings out of 182.

3 What the sources can see

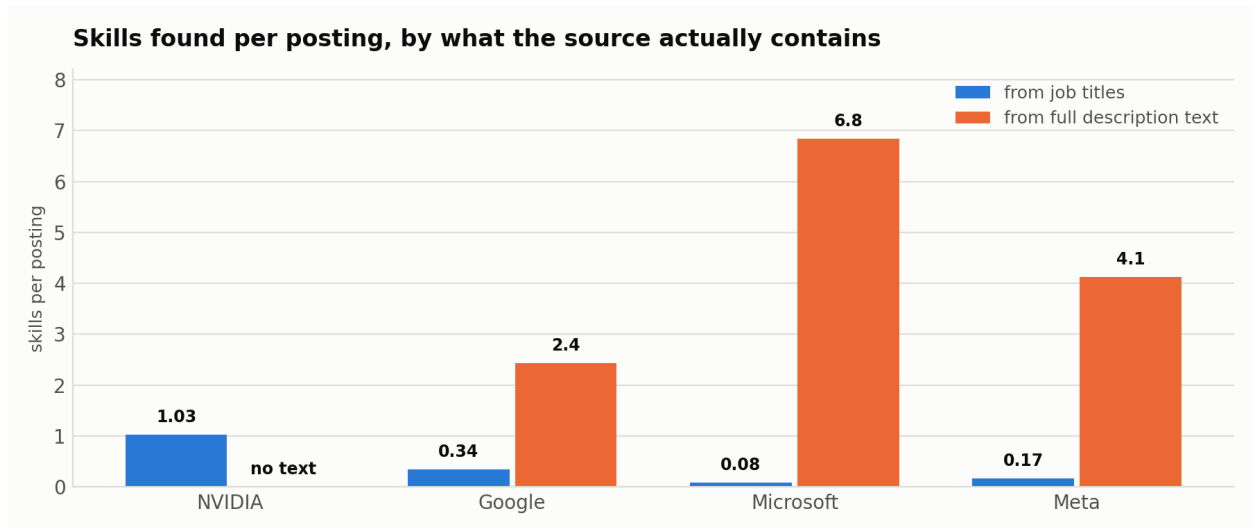


Figure 2. Skills found per posting, by what the source contains.

Company	Postings	Skills per posting, titles	Skills per posting, full text	Coverage, full text
NVIDIA	1,745	1.03	no text available	-
Google	1,227	0.34	2.43	80.5%
Meta	209	0.17	4.13	96.1%
Microsoft	182	0.08	6.84	98.9%

Microsoft's descriptions yield 6.84 skills per posting against 0.08 from its titles, a factor of 85. That single number is the strongest argument in this project for the team decision flagged since Task 2: either Tasks 3, 4, 6 and 8 all run on titles, or the members with text are analysing a different problem from the member without it.

It also shows why NVIDIA's titles are unusual. NVIDIA writes *Senior Physical Design Engineer* and *Senior Mixed Signal Circuit Design Engineer*: the specialisation is in the title. Microsoft writes titles that carry almost no technical content at all, which is why its title-basis coverage is 5.5%.

4 Skill mix, titles basis

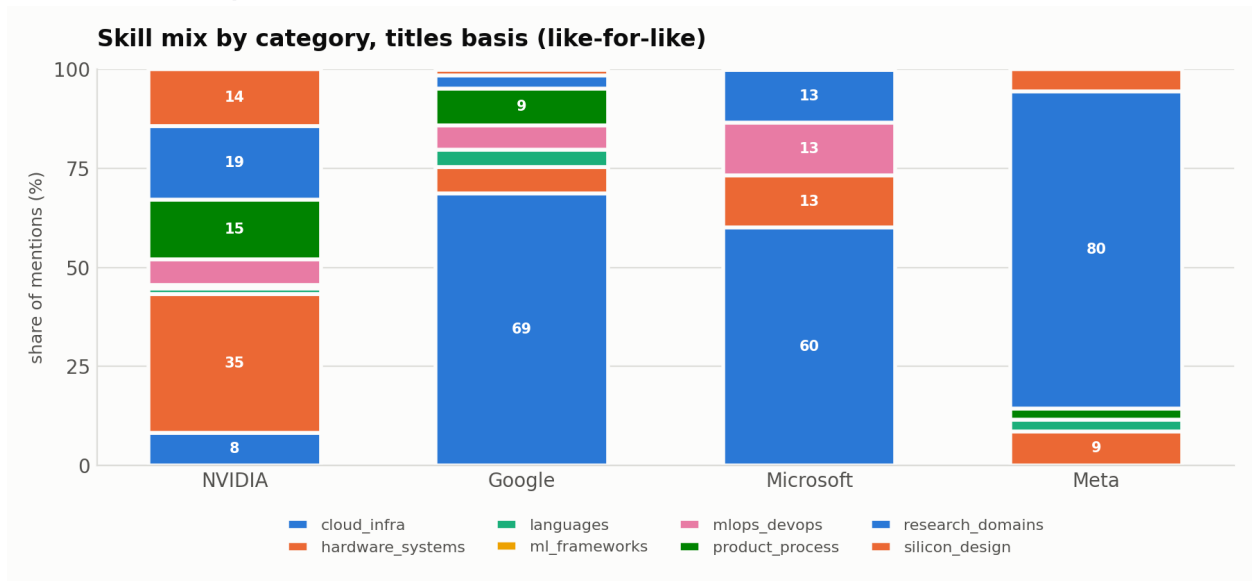


Figure 3. Category mix on the like-for-like titles basis. Read with the small-sample warning below.

Category	NVIDIA	Google	Microsoft	Meta
hardware_systems	35.1%	6.7%	13.3%	8.6%
research_domains	18.6%	3.3%	13.3%	80.0%
product_process	15.0%	9.3%	0.0%	2.9%
silicon_design	14.3%	1.4%	0.0%	5.7%
cloud_infra	8.2%	68.7%	60.0%	0.0%
mlops_devops	6.6%	6.2%	13.3%	0.0%

Small-sample warning: do not quote the Meta and Microsoft columns

On the titles basis Microsoft produces about 15 skill mentions in total and Meta about 35. Meta's 80% research_domains is roughly 28 mentions and Microsoft's 60% cloud_infra is about 9. Percentages that large on counts that small are noise. Only the NVIDIA and Google columns carry enough mentions - 1,793 and 417 - to be read as mix.

Between the two columns that can be read: NVIDIA's mix is dominated by hardware and silicon, 49.4% of mentions between them, against 8.1% for Google. Google's is dominated by cloud infrastructure at 68.7%, which is largely GCP appearing in its own job titles. The two companies are hiring for almost disjoint things, and that part of the comparison is solid because it survives on both taxonomies.

5 Where NVIDIA differs most

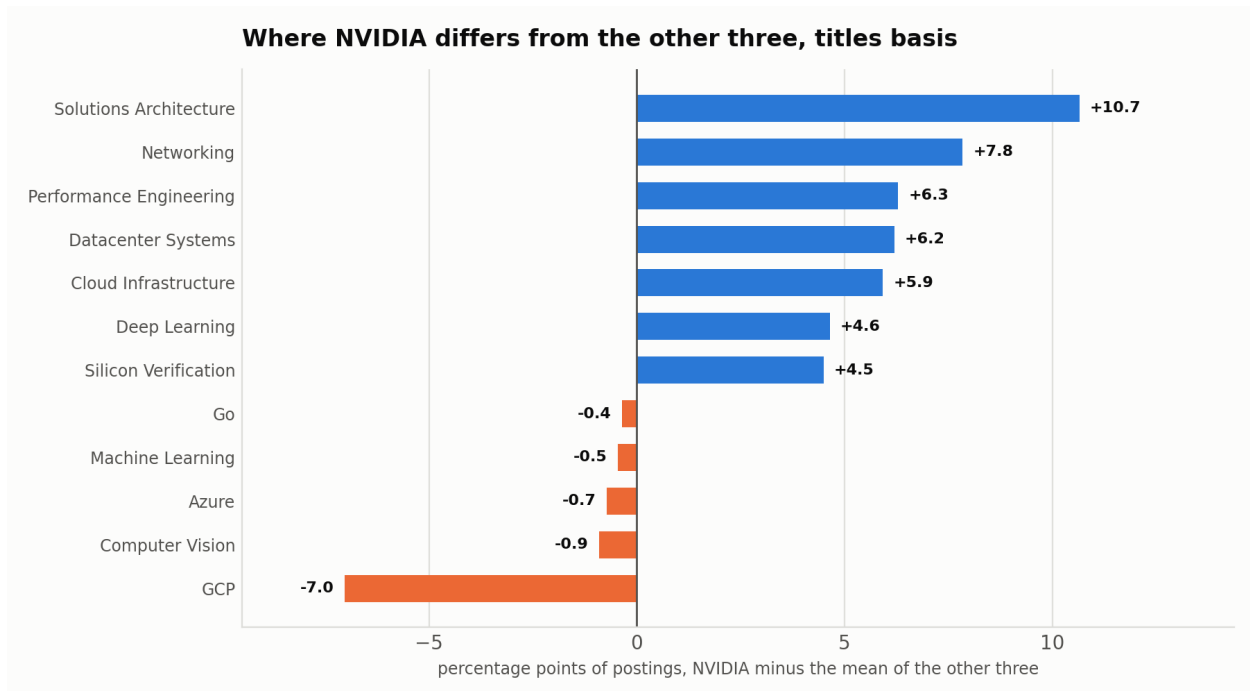


Figure 4. NVIDIA's share of postings minus the mean of the other three, titles basis. Positive means more common at NVIDIA.

The positive side is the silicon and systems stack that no other company in this team hires for: Solutions Architecture +10.7 points, Networking +7.8, Performance Engineering +6.3, Datacenter Systems +6.2, Silicon Verification +4.5, SoC Design +3.5. The negative side is short and almost entirely cloud platforms: GCP -7.0 and Azure -0.7, which is each company advertising its own cloud.

These gaps are inflated by the taxonomy bias in section 2 and should be read as direction, not size. The direction is nonetheless robust: no plausible taxonomy would make Google or Meta look like silicon-design employers.

6 What can and cannot be concluded

#	Conclusion	Confidence
1	NVIDIA hires for hardware, silicon and systems; the other three do not. Hardware plus silicon is 49.4% of NVIDIA's skill mentions against 8.1% for Google.	High. Survives both taxonomies and both bases
2	Each company's own cloud shows up in its own postings - GCP for Google, Azure for Microsoft, neither for NVIDIA.	High. Unambiguous and expected
3	NVIDIA's job titles carry far more technical content than the other three's.	Medium. True on the data, but partly a taxonomy effect
4	The four-way skill-mix comparison in section 4.	Low. Meta and Microsoft columns rest on 15 to 35 mentions
5	Any comparison of hiring velocity, growth or seasonality across the four.	Not possible. Three of four datasets have no usable dates

7 What the team needs to do

1. **Every member extends the taxonomy for their own company** from their own candidate terms, then everyone re-runs Task 4. Until then section 2 applies and the comparison is not valid. This is the single

highest-value action left in the project.

- 2. **Decide titles or descriptions, once, for everyone.** A factor of 85 between Microsoft's two bases is not a detail that can be handled in a footnote.
- 3. **Google and Microsoft need posting dates** if Task 7's forecasting is to mean anything beyond NVIDIA, and Meta's 2024 window needs aligning with the others.
- 4. **Fix the sample sizes.** 182 and 209 postings against 1,745 and 1,227 makes every percentage from Meta and Microsoft fragile.

Files

File	Contents
data/comparison/basis_comparison.csv	Per-company coverage and density on all three bases, including the validity control
data/comparison/skill_share_by_company_titles.csv	Every skill, share of each company's postings, sorted by spread
data/comparison/category_mix_by_company_titles.csv	Category mix per company
data/comparison/nvidia_vs_others_titles.csv	NVIDIA minus the mean of the other three
figures/fig0 ... fig3 .png	The four figures above
compare_companies.py	The comparison, re-runnable from a clean checkout
taxonomy_original_reference.yaml	The pre-extension taxonomy, kept so the control can be reproduced

```
cd work/task-6/nvidia-nayab-khalid
python compare_companies.py
```

Declaration	No personal or sensitive data is present in any output. No new data was collected; the inputs are the four members' committed datasets and the shared taxonomy.
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Date	20 September 2026